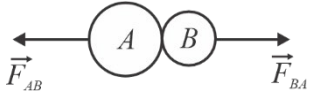
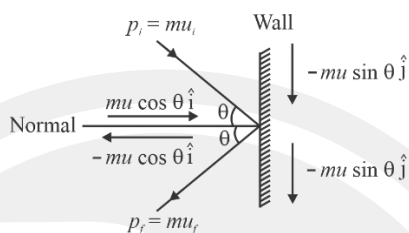
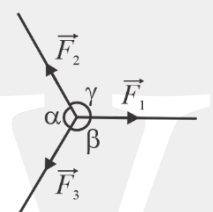
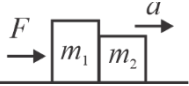
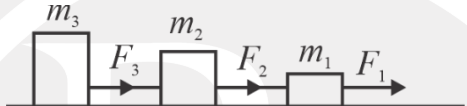
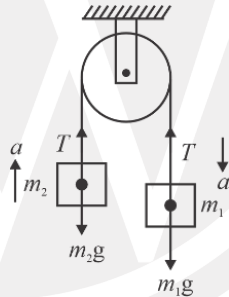
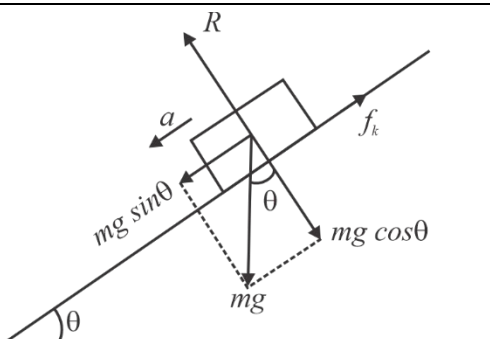
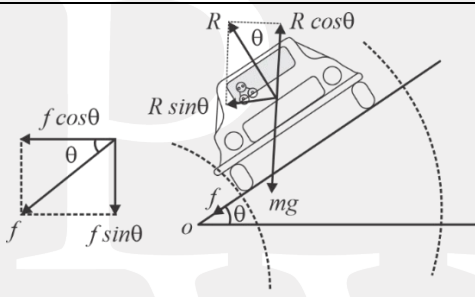


# Laws of Motion

| Sr. No. | Concept                                                     | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Other information                                                                                  |
|---------|-------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| 1.      | Momentum and force                                          | Momentum ( $\vec{p}$ ) = Mass ( $m$ ) $\times$ Velocity ( $\vec{v}$ )<br>$\vec{F} = \frac{d\vec{p}}{dt} = m \frac{d\vec{v}}{dt} = m\vec{a}$                                                                                                                                                                                                                                                                                                                                                                     | $F$ = external force.<br>$dp$ = change in momentum                                                 |
| 2.      | Impulse ( $I$ )                                             | <div>Impulse = Average force <math>\times</math> Time</div> $I = F\Delta t \text{ or } I = \int_{t_1}^{t_2} F dt$ <p>Relation between momentum and impulse:</p> <p>Impulse, <math>I = \int_{t_1}^{t_2} F_{av} dt = F_{av} \int_{t_1}^{t_2} dt = p_2 - p_1</math></p> <p>Graphical method (<math>I</math>) = <math>\int_{t_1}^{t_2} F \cdot dt</math></p>                                                                                                                                                        | $F$ = value of force during impact.<br>$p_2 - p_1$ = change in momentum from time $t_1$ to $t_2$ . |
| 3.      | Conservation of linear momentum                             | $\vec{F}_{ext} = \frac{d\vec{p}}{dt}$ <p>For two particles</p> <p>system <math>\vec{p}_1 + \vec{p}_2 = \text{constant}</math></p> <p>If <math>\vec{F}_{ext} = \vec{0}</math> <span style="margin-left: 100px;"><math>\Delta\vec{p}_1 + \Delta\vec{p}_2 = 0</math></span></p> <p><math>\frac{d\vec{p}}{dt} = \vec{0} \Rightarrow \vec{p} = \text{constant}</math> <span style="margin-left: 50px;"><math>\Delta\vec{p}_1 = -\Delta\vec{p}_2</math></span></p> <p><math>\vec{p}_{in} = \vec{p}_{final}</math></p> | $F_{ext}$ = net external force.<br>$p_{in}$ = initial momentum.<br>$p_{final}$ = final momentum.   |
| 4.      | Action and reaction force<br>(Newton's 3 <sup>rd</sup> law) |  <div><math>\vec{F}_{AB} = -\vec{F}_{BA}</math></div> <p>Force on <math>A</math> by <math>B</math> = - Force on <math>B</math> by <math>A</math></p>                                                                                                                                                                                                                                                                         | $F_{BA}$ = force exerted by body $A$ on $B$<br>$F_{AB}$ = force exerted by $B$ on $A$ .            |

| Sr. No. | Concept                                                        | Formulas                                                                                                                                                                                                                                                                                                                                                                                                           | Other information                                                                                                                                                                                                                        |
|---------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5.      | <b>Newton's 3<sup>rd</sup> law and momentum conservation</b>   | <p>According to Newton's 3<sup>rd</sup> law of motion,</p> $F_{12} = -F_{21}$ $\Rightarrow \frac{m_1(v_1 - u_1)}{t} = -\frac{m_2(v_2 - u_2)}{t}$ <p>or <math>m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2</math></p>                                                                                                                                                                                                          | <p><math>m_1</math> and <math>m_2</math> = masses of the two bodies.</p> <p><math>u_1</math> and <math>u_2</math> = initial velocities of two bodies.</p> <p><math>v_1</math> and <math>v_2</math> = final velocities of two bodies.</p> |
| 6.      | <b>For oblique collision momentum change</b>                   | <p><math>\vec{p}_i = mu \cos \theta \hat{i} - mu \sin \theta \hat{j}</math></p> <p><math>\vec{p}_f = -mu \cos \theta \hat{i} - mu \sin \theta \hat{j}</math></p>  <p>Now, change in momentum, <math>\Delta \vec{p} = \vec{p}_f - \vec{p}_i</math></p> $= -2mu \cos \theta \hat{i}$                                               | <p><math>m</math> = mass of ball.</p> <p><math>u</math> = velocity of the ball.</p> <p><math>\vec{p}_f</math> and <math>\vec{p}_i</math> = final and is the initial momentum.</p>                                                        |
| 7.      | <b>Equilibrium of concurrent forces</b>                        | <p>If <math>\vec{F}_1</math>, <math>\vec{F}_2</math> and <math>\vec{F}_3</math> are three concurrent forces in equilibrium</p> <p>(i) <math>\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}</math></p> <p>(ii) <math>\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}</math></p>  <p>(Lami's theorem)</p> | <p><math>\alpha, \beta, \gamma</math> = opposite angles of the forces.</p>                                                                                                                                                               |
| 8.      | <b>Rocket propulsion</b>                                       | <p>If rocket is accelerating upwards:</p> $v_{rel} \frac{dm}{dt} - mg = ma$ <p>The acceleration of the rocket is given as:</p> $a = \frac{v_{rel}}{m} \frac{dm}{dt} - g$                                                                                                                                                                                                                                           | <p><math>m</math> = mass of rocket</p> <p><math>v_{rel}</math> = velocity of exhaust gases w.r.t rocket</p> <p><math>\frac{dm}{dt}</math> = rate of burning of fuel</p>                                                                  |
| 9.      | <b>The apparent weight of a man in a lift (<math>W</math>)</b> | <p>(i) When the lift moves upwards</p> $W = m(g + a)$ <p>(ii) When the lift moves downwards</p> $W = m(g - a)$ <p>(iii) When the lift falls freely, <math>a = g</math>, so</p> $W = m(g - a) = m(g - g) = 0$ <p>(iv) When the lift is at rest or moves with uniform velocity, <math>a = 0</math>, so <math>W = m(g - 0) = mg</math></p>                                                                            | <p><math>W</math> = normal reaction or apparent weight of the man.</p> <p><math>a</math> = acceleration of the lift or man</p> <p><math>m</math> = mass of the man.</p>                                                                  |

| Sr. No. | Concept                                          | Formulas                                                                                                                                                                                                                                                                                                                         | Other information                                                                                                              |
|---------|--------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------|
| 10.     | Spring force                                     | Spring force $F = -kx$<br>For series combination of springs:<br>$\frac{1}{k_{eq}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$<br>For parallel combination of springs:<br>$k_{eq} = k_1 + k_2 + k_3 + \dots$                                                                                                                         | $x$ = displacement of the free end from its natural length or deformation of the spring.<br>$k$ = spring constant.             |
| 11.     | Two bodies are in contact                        | $a = \frac{F}{(m_1 + m_2)}$<br><br>Contact force on $m_1 = m_1 a = \frac{m_1 F}{(m_1 + m_2)}$<br>Contact force on $m_2 = m_2 a = \frac{m_2 F}{(m_1 + m_2)}$                                                                                    | $F$ = external force acting on a body of mass $m_1$ .<br>$a$ = acceleration of the system.                                     |
| 12.     | Three masses placed on a smooth horizontal table | <br>$a = \frac{F_1}{(m_1 + m_2 + m_3)}, F_2 = \frac{(m_2 + m_3)F_1}{m_1 + m_2 + m_3}$<br>$F_3 = \frac{m_3 F_1}{m_1 + m_2 + m_3}$                                                                                                               | $F_1$ = external force applied on mass $m_1$ .<br>$a$ = acceleration of the system.                                            |
| 13.     | Pully mass system                                | $a = \frac{(m_1 - m_2)}{(m_1 + m_2)} g$<br>$T = \frac{2m_1 m_2}{(m_1 + m_2)} g$<br>                                                                                                                                                          | $a$ = acceleration of the system.<br>$T$ = tension in the string.<br>Here, $m_1 > m_2$                                         |
| 14.     | Friction ( $f_s$ )                               | $\mu_s = \frac{f_s^{\max}}{R} \text{ or } f_s^{\max} = \mu_s R$<br>If $\theta$ is the angle of friction, then $\mu_s = \tan \theta$<br>If $\phi$ is the angle of repose, then $\mu_s = \tan \phi$<br>For a body moving on a rough horizontal surface with retardation $a$ :<br>$\mu = \frac{f}{R} = \frac{ma}{mg} = \frac{a}{g}$ | $\mu_s$ = coefficient of limiting friction.<br>$f_s^{\max}$ = force of limiting friction.<br>$g$ = gravitational acceleration. |

| Sr. No. | Concept                                   | Formulas                                                                                                                                                                                                                                                                                                                                                                                                                                        | Other information                                                                                                                            |
|---------|-------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------|
| 15.     | Motion along rough inclined plane         |  <p>Normal reaction, <math>R = mg \cos \theta</math><br/> Friction, <math>f = \mu R = \mu mg \cos \theta</math><br/> Net force on body down the inclined plane:<br/> <math>F = mg \sin \theta - f = mg (\sin \theta - \mu \cos \theta)</math><br/> Acceleration of the body sliding down:<br/> <math>a = g (\sin \theta - \mu_k \cos \theta)</math></p>       | $\theta$ = angle of inclination.<br>$\mu$ = coefficient of limiting friction.<br>$W$ = work done.<br>$\mu_k$ = coefficient kinetic friction. |
| 16.     | Banking of roads and bending of a cyclist |  <p>Maximum velocity with which the vehicle can safely take a circular turn: <math>v_{\max} = \sqrt{\mu rg}</math><br/> Banking of roads maximum safe velocity:<br/> <math display="block">v_{\max} = \sqrt{rg \left( \frac{\mu + \tan \theta}{1 - \mu \tan \theta} \right)}</math><br/> Bending of a cyclist: <math>\tan \theta = \frac{v^2}{rg}</math></p> | $\mu$ = coefficient of static friction.<br>$v_{\max}$ = maximum velocity of each case.<br>$\theta$ = bending angle                           |
| 17.     | Centripetal force ( $F$ )                 | For a body moving along a horizontal circular path: $F = \frac{mv^2}{r} = mr\omega^2 = mr(2\pi v)^2 = mr \left( \frac{2\pi}{T} \right)^2$                                                                                                                                                                                                                                                                                                       | $v$ = velocity of the body.<br>$T$ = time period.<br>$\omega$ = angular velocity.                                                            |